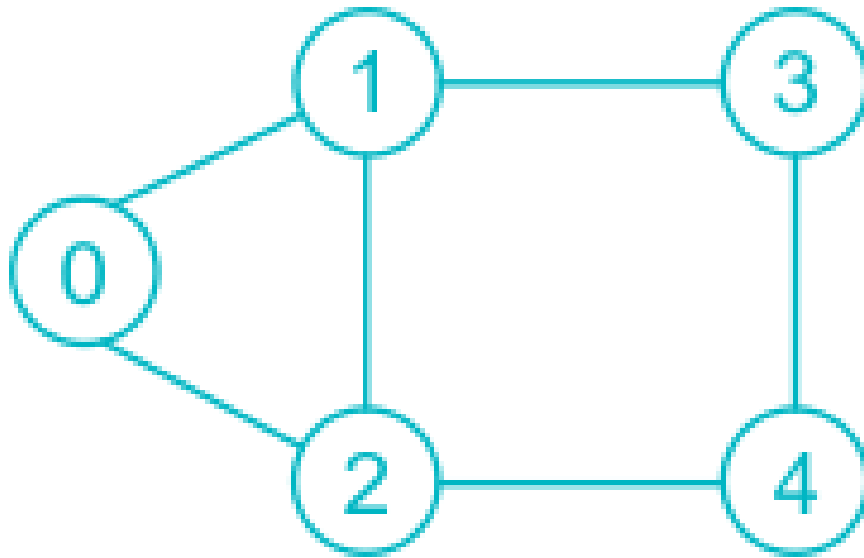
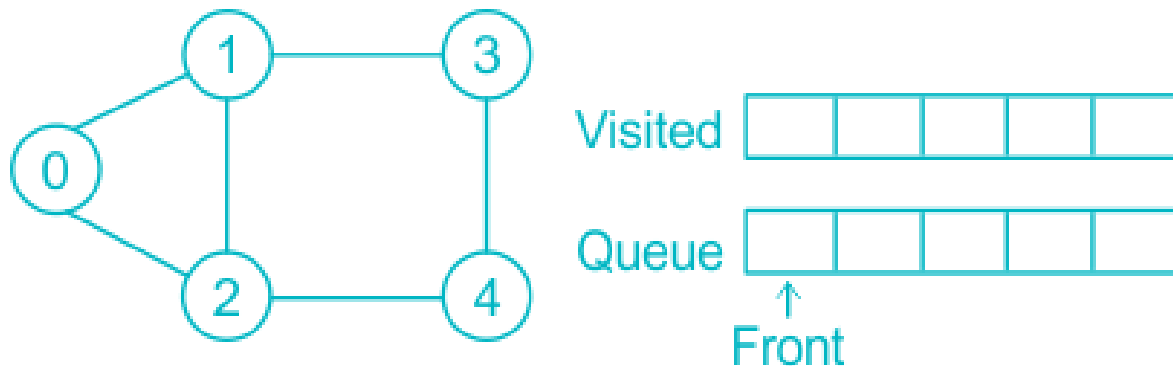


- Please perform a Breadth-First-Search on the following undirected graph. Please show the status of the queue at each iteration. The source is 0.



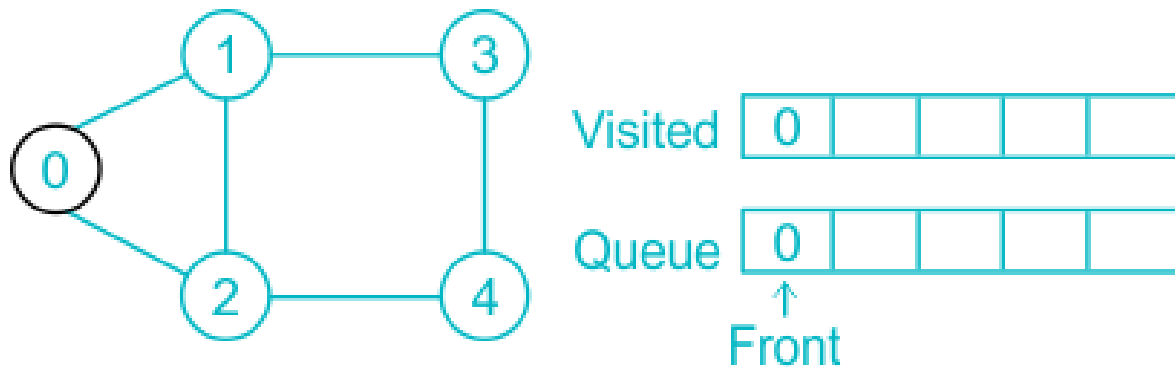
Step 1: Initially the queue and visited arrays are empty.



BFS on Graph

$Q = \{\}$

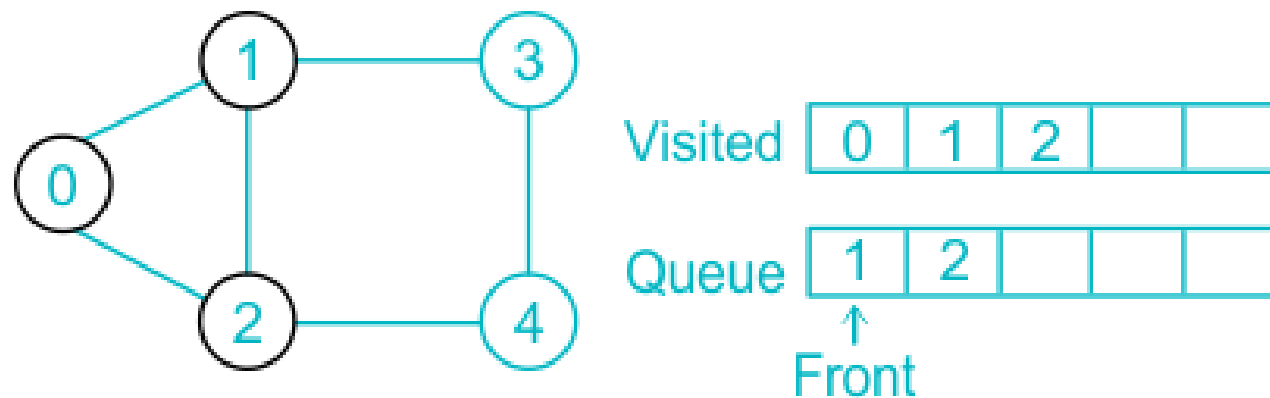
Step 2: Push node 0 into the queue and mark it visited.



BFS on Graph

$Q = \{\}$ $\rightarrow$ $Q = \{0\}$

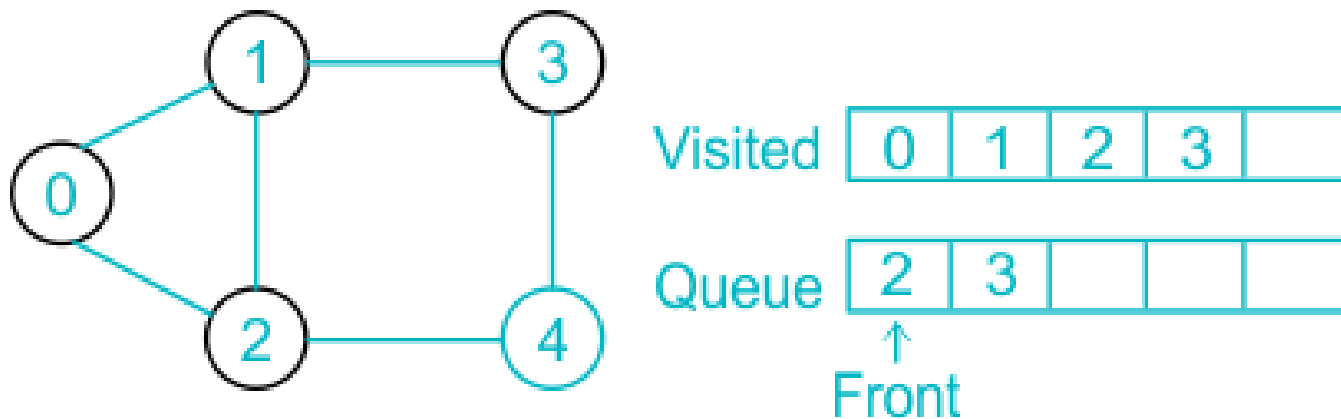
Step 3: Remove node 0 from the front of the queue and visit the unvisited neighbours and push them into the queue.



BFS on Graph

$Q = \{0\} \rightarrow Q = \{1, 2\}$

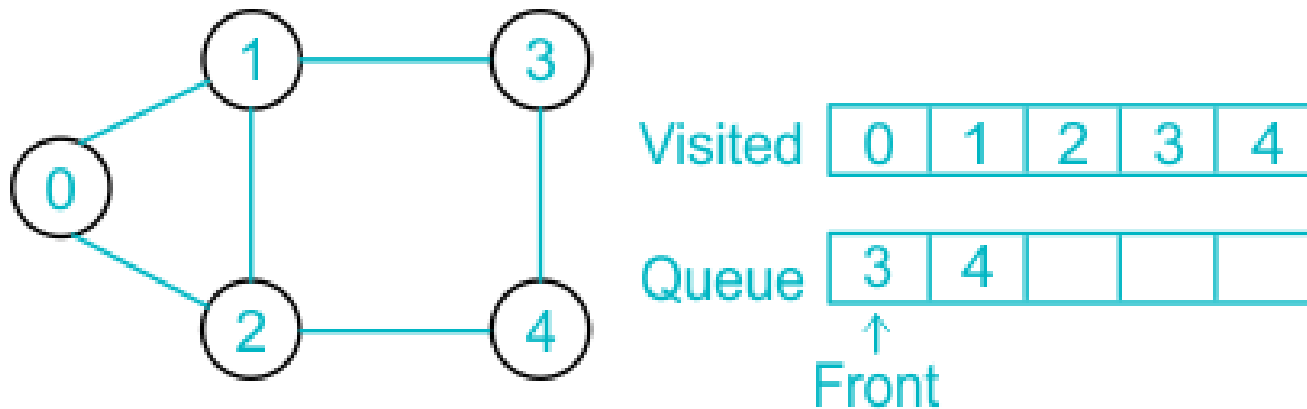
Step 4: Remove node 1 from the front of the queue and visit the unvisited neighbours and push them into the queue.



BFS on Graph

$$Q = \{1, 2\} \rightarrow Q = \{2, 3\}$$

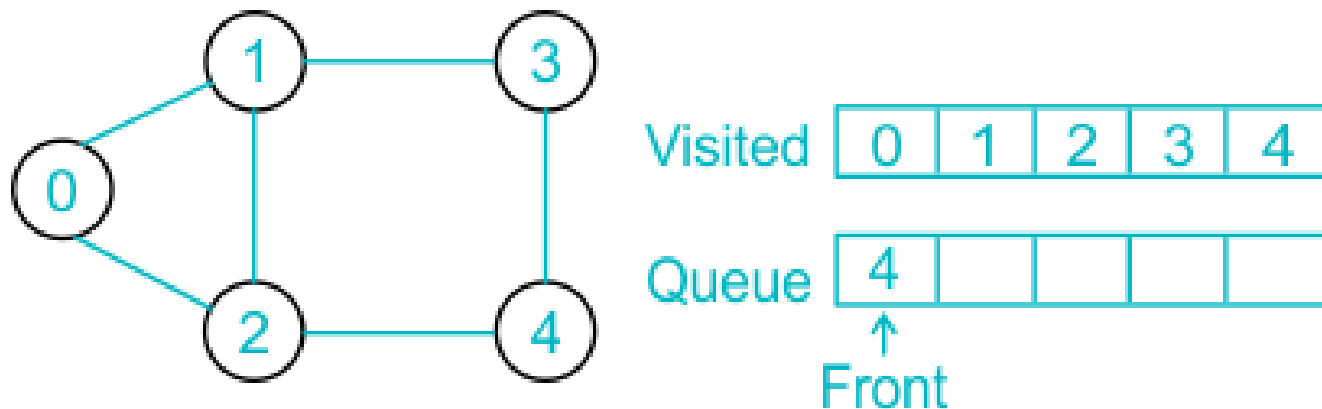
Step 5: Remove node 2 from the front of the queue and visit the unvisited neighbours and push them into the queue.



BFS on Graph

$$Q = \{2, 3\} \rightarrow Q = \{3, 4\}$$

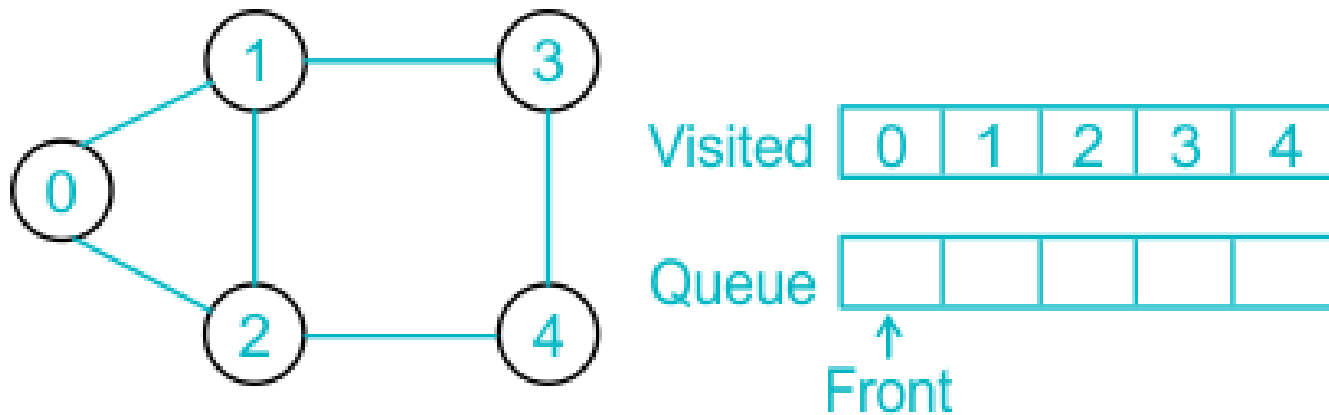
Step 6: Remove node 3 from the front of the queue and visit the unvisited neighbours and push them into the queue.



BFS on Graph

$Q = \{3, 4\} \rightarrow Q = \{4\}$

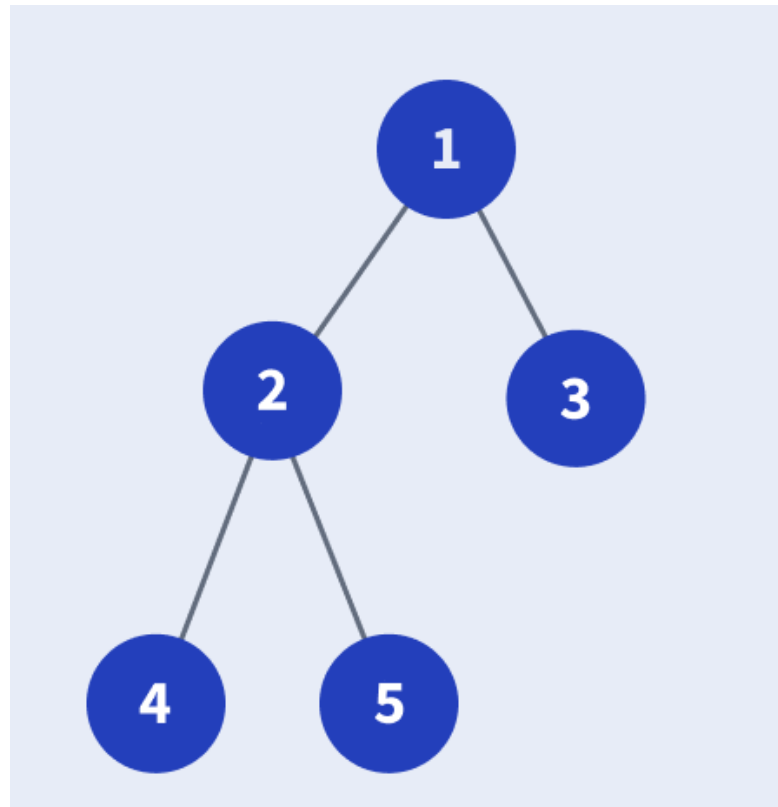
Step 7: Remove node 4 from the front of the queue and visit the unvisited neighbours and push them into the queue.



BFS on Graph

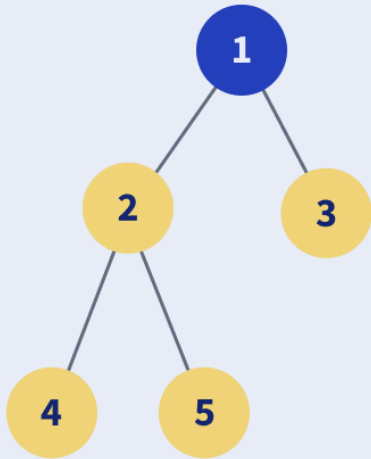
$Q = \{4\} \rightarrow Q = \{\}$

- Please perform a Depth-First-Search on the following undirected graph and show each step of this visit. The source is 1.

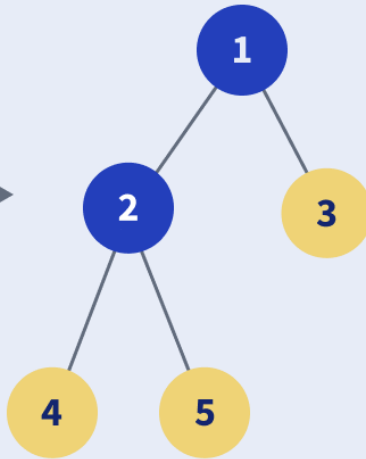


DFS

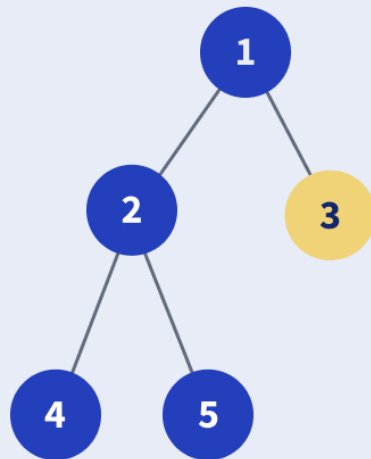
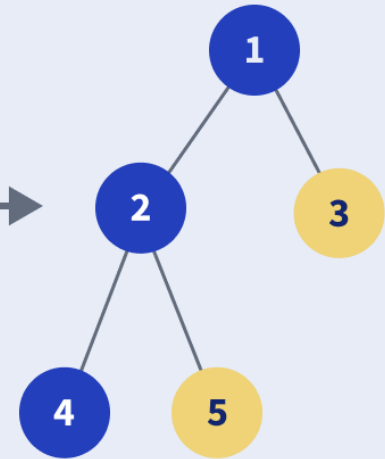
1



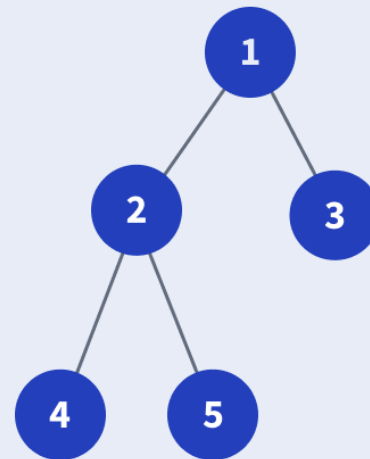
1 → 2



1 → 2 → 4



1 → 2 → 4 → 5



1 → 2 → 4 → 5 → 3